

## Causal attention

|     | The | cat | sat | on | the |
|-----|-----|-----|-----|----|-----|
| The | 1   | 0   | 0   | 0  | 0   |
| cat | 1   | 1   | 0   | 0  | 0   |
| sat | 1   | 1   | 1   | 0  | 0   |
| on  | 1   | 1   | 1   | 1  | 0   |
| the | 1   | 1   | 1   | 1  | 1   |

Each query attends to **every** previous token and itself — lower-triangular mask.

## Attention + sink token

|     | The | cat | sat | on | the |
|-----|-----|-----|-----|----|-----|
| The | 1   | 0   | 0   | 0  | 0   |
| cat | 1   | 1   | 0   | 0  | 0   |
| sat | 1   | 1   | 1   | 0  | 0   |
| on  | 1   | 1   | 1   | 1  | 0   |
| the | 1   | 1   | 1   | 1  | 1   |

↑  
sink token

**Every** query also attends the first (sink) token — the softmax dumps spare mass there.

$\text{softmax}([q \cdot k_0 \cdots q \cdot k_i, s])$  —  $s$  is a learned per-head sink logit (no value vector)

attended (1)     sink column (always on)     masked (0)